Introduction to Statistical Learning and Application - Master of Applied Mathematics

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February 24, 2026

Outline

1. Classification
2. Logistic Regression
3. Evaluation metrics
4. Lab: Classifcation Methods

Motivation

The linear regression model assumes that the response variable Y is quantitative.

- Predicting a qualitative response for an observation can be referred to as **classifying** that observation, since it involves assigning the observation to a category, or class.
- On the other hand, often the methods used for classification first **predict the probability** that the observation belongs to each of the categories of a qualitative variablen.

Why not regression?

1. A regression method cannot accommodate a qualitative response with more than two classes;
2. A regression method will not provide meaningful estimates of $Pr(Y|X)$, even with just two classes.

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What is Classification?

Definition

Classification is a type of supervised machine learning where the goal is to categorize input data into predefined classes or labels.

Types of Classification Algorithms

1. Linear Classifiers:
   - Logistic Regression
   - Support Vector Machines (SVM)
2. Nearest Neighbor Classifiers:
   - k-Nearest Neighbors (k-NN)
3. Tree-Based Classifiers:
   - Decision Trees
   - Gradient Boosting Machines (e.g., XGBoost)
4. Naive Bayes Classifiers

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The `Default` dataset (credit risk example)

We study a binary classification problem: default $\in \{\text{Yes}, \text{No}\}$
The goal is to predict whether an individual will default on their credit card payment.

- Default: missing a required credit card payment
- Balance: average monthly *credit card debt* (not bank money)
- Income: annual income of the individual

Key observations

- High balances are strongly associated with default
- Income alone is a weak predictor
- Classes overlap $\Rightarrow$ no perfect separation

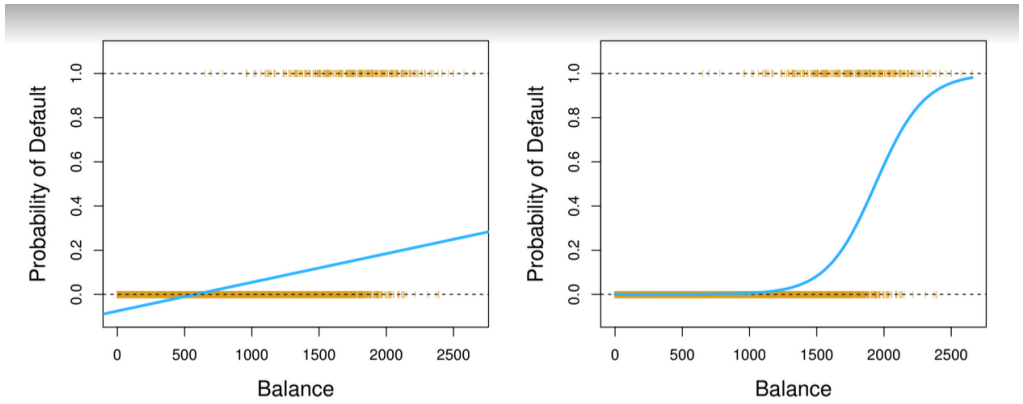
Logistic Regression

Key Note

Rather than modeling this response Y directly, logistic regression models the probability that Y belongs to a particular category.

$$p(X) = \beta_0 + \beta_1(X) \tag{1}$$

Example: For the Default data, we model Pr(default = Yes|balance).



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The Logistic Model

Problem

For balances close to zero, we predict a negative probability of default; if we were to predict for very large balances, we would get values bigger than 1.

To avoid this problem, we must model p(X) using a function that gives outputs between 0 and 1 for all values of X. Many functions meet this description. In logistic regression, we use the logistic function,

$$p(X) = \frac{e^{\beta_0+\beta_1 X}}{1 + e^{\beta_0+\beta_1 X}} \tag{2}$$

The Logistic Model

- After a bit of manipulation of (2), we find that:

$$\frac{p(X)}{1-p(X)} = e^{\beta_0+\beta_1 X} \tag{3}$$

- The left-hand side is called the odds.
- By taking the logarithm of both sides of (3), we arrive at

$$\log\left(\frac{p(X)}{1-p(X)}\right) = \beta_0 + \beta_1 X.$$

- The left-hand side is called the log odds or logit.

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Estimating the Regression Coefficients- The method of maximum likelihood

The basic intuition behind using maximum likelihood

We seek estimates for β_0 and β_1 such that the predicted probability $P(\hat{xi})$ of default for each individual, using (2), corresponds as closely as possible to the individual's observed default status.

This intuition can be formalized using a mathematical equation called a likelihood function:

$$\ell(\beta_0, \beta_1) = \prod_{i:y_i=1} p(x_i) \prod_{i':y_{i'}=0} (1 - p(x_{i'})).$$

Why Maximum Likelihood for Logistic Regression?

Key idea (high level)

In logistic regression, we model probabilities of a binary outcome, not the outcome itself.

- In linear regression: coefficients are estimated by least squares (closed form).
- In logistic regression: $y_i \in \{0, 1\}$ and $\hat{p}_i$ must be in $[0, 1]$.
- We choose coefficients β that make the **observed labels** most plausible under the model.

$$p_i = \mathbb{P}(Y_i = 1 \mid x_i) = \sigma(\eta_i), \quad \eta_i = \beta_0 + \beta^\top x_i, \quad \sigma(z) = \frac{1}{1 + e^{-z}}.$$

From Individual Probabilities to a Likelihood

- For each observation i:
  - if $y_i = 1$, the model assigns probability p_i
  - if $y_i = 0$, the model assigns probability $1 - p_i$
- The likelihood multiplies contributions across all observations.

Likelihood of the full sample

$$L(\beta) = \prod_{i=1}^{n} p_i^{y_i} (1 - p_i)^{1 - y_i}$$

- Maximizing $L(\beta)$ means: predicted probabilities match observed outcomes as well as possible.

Why Use the Log-Likelihood?

- Products of many probabilities can be numerically unstable.
- Taking logs converts products into sums (easier to optimize).
- Maximizing $L(\beta)$ is equivalent to maximizing $\ell(\beta) = \log L(\beta)$.

Log-likelihood

$$\ell(\beta) = \sum_{i=1}^{n} \left[y_i \log(p_i) + (1 - y_i) \log(1 - p_i) \right]$$

- Intuition: each observation contributes a reward/penalty depending on how well it is predicted.

How Are the Coefficients Found in Practice?

Important practical point

Unlike linear regression, logistic regression does **not** have a simple closed-form solution for $\hat{\beta}$.

- We solve:

$$\hat{\beta} = \arg\max_{\beta} \ \ell(\beta)$$

- The log-likelihood is smooth and (for standard logistic regression) concave $\Rightarrow$ a unique optimum.
- Numerical algorithms used in software:
  - Gradient ascent / quasi-Newton (e.g., BFGS)
  - Newton–Raphson / IRLS (Iteratively Reweighted Least Squares)

In practice: R / Python / MATLAB compute $\hat{\beta}$ automatically.

Interpreting Logistic Regression Coefficients

- Logistic regression models the **log-odds**:

$$\log\left(\frac{p(x)}{1-p(x)}\right) = \beta_0 + \beta^\top x$$

- Interpretation of β_j:
  - a one-unit increase in x_j changes the log-odds by β_j
  - holding other predictors constant
- Exponentiating gives an **odds ratio**: $\exp(\beta_j) =$ multiplicative change in the odds for a 1-unit increase in x_j

Quick intuition

$\beta_j > 0$ increases odds (and usually probability); $\beta_j < 0$ decreases odds.

Summary

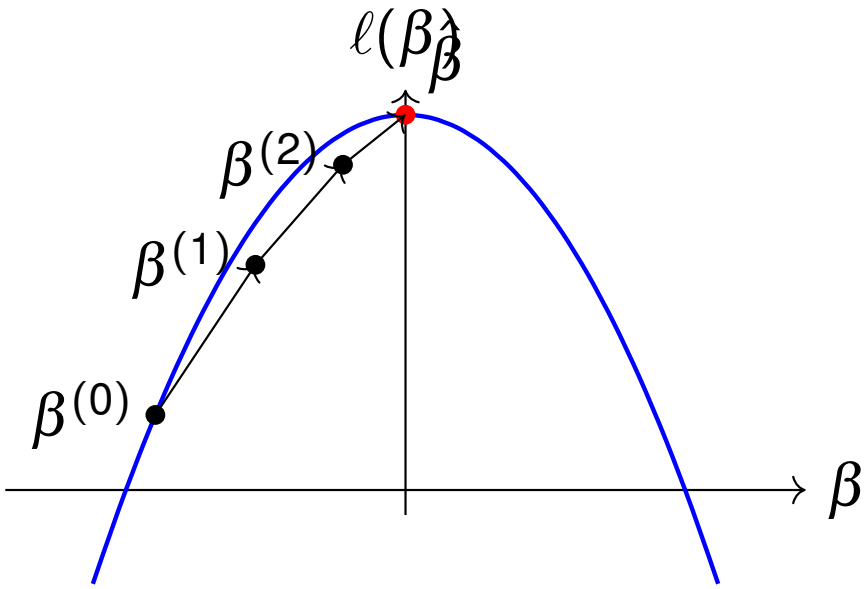
Logistic regression: estimation and meaning

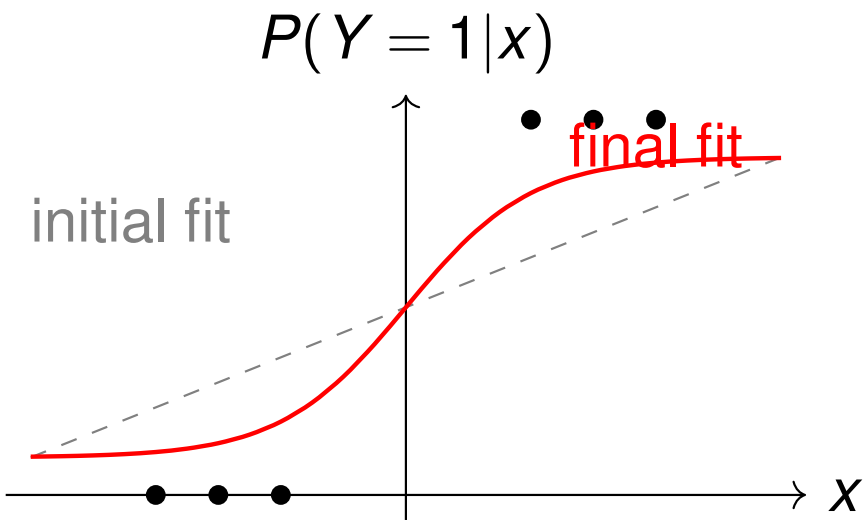
- We model probabilities: $p_i = \mathbb{P}(Y_i = 1 \mid x_i)$.
- Coefficients are estimated by **maximum likelihood**.
- We maximize the log-likelihood:

$$\ell(\beta) = \sum_{i=1}^{n} \Big[y_i \log(p_i) + (1 - y_i) \log(1 - p_i) \Big].$$

- Optimization is numerical (IRLS / Newton methods).
- Coefficients are interpreted via log-odds and odds ratios.

Visual Intuition





Example SEstimating the Regression Coefficients

Table 1 shows the coefficient estimates and related information that result from fitting a logistic regression model on the Default data in order to predict the probability of default=Yes using balance.

	Coefficient	Std. error	z-statistic	p-value
`Intercept`	-10.6513	0.3612	-29.5	<0.0001
`balance`	0.0055	0.0002	24.9	<0.0001

For the Default data, estimated coefficients of the logistic regression model that predicts the probability of default using balance. A one-unit increase in balance is associated with an increase in the log odds of default by 0.0055 units.

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Multiple Logistic Regression

By analogy with the extension from simple to MLR , we can generalize (4) as follows, where X=($X_1, \ldots,_p$) are p predictors.

$$\log \frac{\mathrm{p(X)}}{1-\mathrm{p(X)}} = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

Table 2 shows the coefficient estimates that uses balance, income, and student status to predict probability of default.

	Coefficient	Std. error	z-statistic	p-value
`Intercept`	−10.8690	0.4923	−22.08	<0.0001
`balance`	0.0057	0.0002	24.74	<0.0001
`income`	0.0030	0.0082	0.37	0.7115
`student[Yes]`	−0.6468	0.2362	−2.74	0.0062

Student status is encoded as a dummy variable student[Yes], with a value of 1 for a student.

Interpreting Coefficients in Multiple Logistic Regression

- The model describes changes in **log-odds**

$$\log\left(\frac{p(X)}{1-p(X)}\right)$$

Interpretationof a single coefficient β_j

A one-unit increase in X_j changes the log-odds of $Y = 1$ by β_j, **holding all other predictors fixed**.

- This is different from simple logistic regression
- Effects may change sign or significance when new variables are added

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Beyond Binary Logistic Regression

- Logistic regression handles $Y \in \{0, 1\}$
- Many problems involve more than two classes:
  - disease type
  - risk level
  - object category
- We need a model for $Y \in \{1, \ldots, K\}$

Solution

Multinomial (multiclass) logistic regression

Reference Class Idea

- One class is chosen as a baseline (reference)
- Without loss of generality, choose class K
- All other classes are compared to this reference

Why do we need a reference?

- To avoid redundancy
- To ensure model identifiability

Multinomial Logistic Regression Model

For $k = 1, \ldots, K-1$:

$$\Pr(Y = k \mid X = x) = \frac{e^{\beta_{k0} + \beta_k^\top x}}{1 + \sum_{\ell=1}^{K-1} e^{\beta_{\ell 0} + \beta_\ell^\top x}}$$

For the reference class:

$$\Pr(Y = K \mid X = x) = \frac{1}{1 + \sum_{\ell=1}^{K-1} e^{\beta_{\ell 0} + \beta_\ell^\top x}}$$

Key property

Probabilities are positive and sum to 1

Interpretation via Log-Odds

For $k = 1, \ldots, K-1$:

$$\log\left(\frac{\Pr(Y = k \mid X = x)}{\Pr(Y = K \mid X = x)}\right) = \beta_{k0} + \beta_k^\top x$$

Interpretation

- Each class has its own regression equation
- Coefficients compare class k to the reference class
- Effects are interpreted **holding other predictors fixed**

Example

- *Consider an emergency room setting with three possible medical conditions: stroke, drug overdose, and epileptic seizure.*
- *Multinomial logistic regression requires choosing one category as a baseline (reference) class. For example, we may choose stroke or drug overdose as the baseline.*
- *Changing the baseline affects the numerical values of the coefficients, but **does not affect**:*
  - *the predicted probabilities,*
  - *the log-odds between any pair of classes,*
  - *or the overall model fit.*
- *Therefore, coefficient interpretation must always be made relative to the chosen baseline.*

Multinomial Logistic Regression: Softmax Coding

- In softmax coding, **no reference class is chosen**.
- All K classes are treated **symmetrically**.
- Each class k is assigned a linear score:

$$\eta_k(x) = \beta_{k0} + \beta_{k1}x_1 + \cdots + \beta_{kp}x_p$$

Softmax probabilities

$$\Pr(Y = k \mid X = x) = \frac{e^{\eta_k(x)}}{\sum_{\ell=1}^{K} e^{\eta_\ell(x)}}$$

- The denominator ensures probabilities are positive and sum to 1.

Interpretation of Softmax Coefficients

Log-odds between two classes

$$\log\left(\frac{\Pr(Y=k \mid X=x)}{\Pr(Y=k' \mid X=x)}\right) = (\beta_{k0} - \beta_{k'0}) + (\beta_k - \beta_{k'})^\top x$$

- Interpretation is always relative between two classes.
- Individual coefficients β_k have no standalone meaning.
- Only differences between coefficients matter.

Key message

Softmax coding and baseline coding give the same predictions; they are just two equivalent parameterizations.

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Confusion Matrix: the foundation

Binary classification outcomes

	Predicted 0	**Predicted 1**
Actual 0	TN	FP
Actual 1	FN	TP

- TP: true positives (correctly detected positives)
- FP: false positives (false alarms)
- FN: false negatives (missed positives)
- TN: true negatives

Accuracy

Definition

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

- Measures overall correctness
- Easy to interpret

Limitation

Accuracy can be misleading when classes are imbalanced.

`sklearn.metrics.accuracy_score`

Precision and Recall

Definitions

$$\text{Precision} = \frac{TP}{TP + FP} \qquad \text{Recall} = \frac{TP}{TP + FN}$$

- **Precision**: How many predicted positives are correct?
- **Recall**: How many actual positives are detected?

Interpretation

- High precision $\Rightarrow$ few false alarms
- High recall $\Rightarrow$ few missed detections

`precision_score`, `recall_score`

F1-score

Definition

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

- Harmonic mean of precision and recall
- Penalizes extreme imbalance between the two

When to use F1-score

When both false positives and false negatives matter.

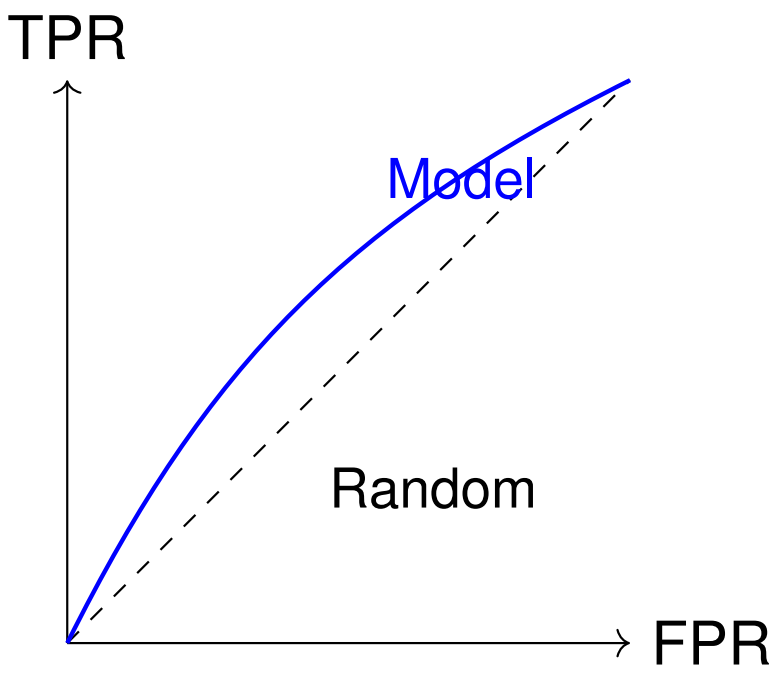
`sklearn.metrics.f1_score`

ROC Curve

Key idea

The ROC curve evaluates performance across **all decision thresholds**.

- TPR = Recall = $\frac{TP}{TP+FN}$
- FPR = $\frac{FP}{FP+TN}$
- Each point = one threshold



AUC: Area Under the Curve

Definition

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR})\, d(\text{FPR})$$

- Threshold-independent metric
- Probability that a random positive is ranked above a random negative

Interpretation

- AUC = 0.5 â random guessing
- AUC = 1.0 â perfect classifier

`sklearn.metrics.roc_auc_score`

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   - Gaussian Naive Bayes
   - K-Nearest Neighbors (KNN)

Lab 1: Social_Network_Ads (Logistic Regression)

Objective

Build a logistic regression classifier to predict whether a user purchases a product (`Purchased` $\in \{0, 1\}$) using two predictors: `Age` and `EstimatedSalary`.

The dataset contains user profiles from a social network advertisement campaign. Each row corresponds to one user.

Instructions

- Open a Python notebook (Jupyter or Google Colab).
- Create one section per step of the lab (Step 1 till 5).
- For each step: include your code, the requested plots/tables, and short answers to the questions.

Deadline: March 10

Step 1: Load & Inspect the Dataset

- Load the CSV file and inspect the table structure
- Identify predictors (`Age`, `EstimatedSalary`) and target (`Purchased`)
- Check missing values and class balance

Questions

- How many observations and variables?
- Is the target balanced?
- Any missing values?

Python hints

`pd.read_csv`, `df.head()`, `df.info()`, `value_counts()`, `isna().sum()`

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Step 2: Visualize the Classes (Separability?)

- Create a scatter plot in feature space: `Age` (x-axis) vs `EstimatedSalary` (y-axis)
- Color points by `Purchased` (0/1)

Questions

- Do the classes appear (approximately) linearly separable?
- What would a linear decision boundary look like here?

Python hints

- `import matplotlib.pyplot as plt`
- `plt.scatter(x, y, c=labels)`
- Color options such as `cmap="bwr"` and transparency `alpha=0.7`
- Axis labels with `plt.xlabel()` and `plt.ylabel()`

Step 3: Fit Logistic Regression (Train/Test)

- Split data into training and test sets
- Fit logistic regression on (`Age`, `EstimatedSalary`)
- Extract coefficients and interpret signs

Questions

- What are the signs of the coefficients?
- Which predictor increases the odds of purchase?

Python hints

- Split the data: `train_test_split` from `sklearn.model_selection`
- Fit the model: `LogisticRegression` from `sklearn.linear_model`
- Access coefficients: `clf.coef_` and `clf.intercept_`

Step 4: Predictions and Model Evaluation

- Use the trained model to predict:
  - class labels (0/1)
  - predicted probabilities
- Evaluate how well the model performs on the test set

Questions

- What is the difference between predicted probabilities and predicted classes?
- How accurate is the model on the test set?
- Is accuracy alone sufficient to evaluate performance?

Python hints

- `clf.predict(X_test)`
- `clf.predict_proba(X_test)`
- `accuracy_score` from `sklearn.metrics`

Step 5: Effect of the Classification Threshold

Logistic regression outputs probabilities in $[0, 1]$. A threshold τ is used to convert probabilities into class labels; We usually fix $\tau = 0.5$, but this is not always optimal

- Recompute confusion matrices for different values of τ

Questions

- What happens if the threshold is lowered (e.g. $\tau = 0.3$)?
- What happens if the threshold is increased (e.g. $\tau = 0.7$)?
- Which types of errors are affected (false positives vs false negatives)?

Python hints

- `clf.predict_proba(X_test)[:,1]`
- Convert to labels: `(proba >= tau).astype(int)`